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Image enhancement algorithm combining histogram equalization and bilateral filtering

Mingzhu Wu [a], Qiuyan Zhong [b,*]

[a] Department of Information Engineering, Guangzhou Institute of Technology, Guangzhou 510075, China
[b] Asset Management Division, Guangzhou Railway Polytechnic, Guangzhou 511308, China

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ABSTRACT

In the process of image acquisition, transmission, and storage, the image quality is often degraded due to a variety of unfavorable factors, resulting in information loss, which poses certain difficulties for subsequent image processing and analysis. How to enhance the visibility of image details and maintain the naturalness of the image is one of the important challenges in image processing. In response to this challenge, an image enhancement algorithm is proposed based on the advantages of histogram equalization and bilateral filtering. This algorithm organically integrates histogram equalization and bilateral filtering, aiming to improve image quality while reducing noise in the image. Specifically, the study first utilizes an improved histogram equalization strategy to preprocess the image and then applies a bilateral filter for further optimization. The experimental results showed that the optimized histogram equalization could effectively improve the global contrast of the image and avoid excessive enhancement and gray phenomenon of the image. Moreover, its peak signal-to-noise ratio could reach 0.71. However, bilateral filters showed significant advantages in processing complex data sets, and the peak signal-to-noise ratio could reach 0.95. It illustrated that the optimal research method has obvious advantages in improving image quality and reducing noise. The new enhancement strategy not only significantly improves the global contrast of the image but also preserves the naturalness of the image, providing important technical support for image analysis, machine vision, and artificial intelligence applications.

1. Introduction

The objective of the introduction part is to elucidate the significance of image enhancement techniques and the constraints of existing techniques. Furthermore, it outlines how the algorithms presented in this study offer solutions to these challenges.

The rapid development of technology has made image enhancement play an important role in medical image processing, satellite remote sensing images, and security monitoring. However, during image acquisition, transmission, and storage, quality degradation often occurs due to various adverse factors, leading to information loss, which poses great challenges for subsequent image processing and analysis. There are currently two main technologies to address this issue: histogram equalization (HE) and bilateral filtering [1,2]. HE can improve image contrast and enhance image details, while bilateral filtering, as a nonlinear filtering technique, can effectively suppress noise while protecting image edge information. Bilateral filtering is a powerful image processing technique that has been widely employed in numerous image enhancement fields, including stereo vision. In this context, it is utilized to enhance the estimation of parallax maps [3]. Although existing techniques such as HE and bilateral filtering have achieved some results in image enhancement, they have their own limitations. For example, HE can result in excessive image enhancement and color distortion. However, bilateral filtering may not be effective when dealing with large uniform areas. In the process of image acquisition, transmission, and storage, image quality is often degraded due to various adverse factors, resulting in information loss, which poses a great challenge to subsequent image processing and analysis. With the limitations of existing techniques and the degradation of image quality, it is necessary to develop a new image enhancement algorithm. This algorithm must improve the global contrast of the image, maintain the naturalness of the image, and effectively suppress the noise. Considering these issues, a novel image enhancement algorithm has been proposed. This algorithm first utilizes HE to enhance image contrast, then uses bilateral filtering to protect image edge information, and then automatically adjusts the enhancement effect based on the characteristics of the image through an

* Corresponding author.
E-mail address: zhong9079@163.com (Q. Zhong).

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adaptive parameter adjustment strategy [4]. This effectively avoids the problems of excessive and insufficient enhancement, improves visual effects and image quality, and provides more accurate information for subsequent image processing and analysis [5,6]. According to the experimental results, this new image enhancement algorithm outperforms existing image enhancement methods on multiple standard datasets. The innovation of the research method is as follows: (1) The HE and bilateral filtering technology are organically combined to achieve the overall improvement of image quality through complementary advantages. (2) The adaptive parameter adjustment strategy is introduced to automatically adjust the enhancement effect according to the image characteristics, effectively avoiding the problem of over-enhancement or under-enhancement. (3) Through the improved HE strategy, the global contrast of the image is significantly improved, and the color distortion and gray-scale phenomenon are avoided. The research contributions are as follows: (1) The research presents a novel image enhancement technology scheme that offers technical support for the implementation of image analysis, machine vision, and artificial intelligence. (2) This study offers a novel approach and methodological framework for the field of image enhancement, which considers the global and local characteristics of images in a comprehensive manner. (3) The proposed algorithm has significant practical value and popularization potential in practical applications, such as medical image processing and satellite remote sensing image analysis. The research will be carried out in five parts. The first part is an overview of the construction and method research of image enhancement algorithms that integrate HE and bilateral filtering. The second part is a model for the construction and method research of image enhancement algorithms that integrate HE and bilateral filtering. The third part is an experimental verification. The fourth part is a discussion of the results. The fifth part is a summary.

2. Literature review

The related works part is designed to review and analyze the current research in the field of image enhancement, providing theoretical and practical background for understanding the algorithm proposed in this study.

As the demand for image processing has grown, numerous scholars have come to recognize the significance of image enhancement technology. The global HE algorithm is designed to achieve image enhancement by counting the brightness of image pixels and stretching the histogram. Some scholars have carried out related research on global HE algorithm. Considering the correlation between image quality and environment, Wen H Q et al. proposed an image enhancement algorithm based on multi-threshold and HE. In the process, the low-light image was converted from RGB color space to YUV color space, and the adaptive threshold enhancement was done. Experimental results showed that the proposed method enhanced the brightness and contrast of images [7]. Rahman et al. proposed using image enhancement and lung segmentation technology to improve the detection accuracy of COVID-19. The research results showed that the image enhancement technology based on gamma correction was superior to other technologies. The newly proposed UNet model achieved an intersection to union ratio of 94.3 % and a Dice coefficient of 96.94 % in lung segmentation. The proposed method could realize rapid and reliable detection of COVID-19 and help reduce the burden of the medical system [8]. Bi et al. utilized HE to achieve tamper detection by utilizing trace features. Their method was robust to conventional HE operations, small resolution image equalization operations, and compressed equalization images, which could distinguish between HE and other types of contrast enhancement operations [9]. To expand the application scope of HE method, Jiang JL et al. adopted fuzzy C-means clustering algorithm based on histogram weighting to rapidly cluster images. Experimental results showed that the proposed method had better enhancement performance in image enhancement [10]. To improve the visual quality of images, Paul Abhisek et al. proposed a method of double HE based on histogram correction. During the process, the input image histogram was changed and clipped according to their respective platform limit parameters to control the result of excessive enhancement. The experimental results showed that the proposed method could effectively improve the visual quality of images [11].

Bilateral filtering is a nonlinear filtering algorithm that combines image spatial information and content information. It can solve the shortcomings of wavelet transform in processing images with rich texture information. While eliminating noise and smoothing images, it can also effectively protect edge information. There are also some scholars on bilateral filtering technology related research. Deng et al. proposed an image retrieval method which eliminated noise, smoothes images, and protects edge information. Experiments showed that it had a high signal-to-noise ratio [12]. LUO G Q et al. proposed a technique to improve bilateral filtering for image enhancement. The average correction technique was introduced into the algorithm to enhance the contrast and clarity of the image. Experimental results indicated that the proposed method can run well, but it cannot achieve good results for image processing containing complex noise similar to edge pixels [13]. Jia et al. proposed an improved bilateral filtering algorithm to address these issues. The efficacy of adaptive bilateral filtering can be enhanced by determining whether the pixels at the edge of the image are noisy and adopting a parameter adaptive setting method. Through experimental simulation verification, the proposed algorithm achieved good results in both subjective observation and objective evaluation criteria [14]. Chen et al. proposed a new filter to solve the above problem. They used Gaussian space kernels to obtain low-pass guidance for distance kernels. This provided a clear Gaussian range kernel for future synthesis. Experimental results on multiple test datasets showed that it outperformed other methods based on bilateral filters [15].

In summary, a large number of scholars have studied the image enhancement algorithms of HE and bilateral filtering and proved that the two methods have certain prospects in the field of image enhancement. Bilateral filtering has good edge preservation performance, while the global HE algorithm stretches the histogram to achieve image enhancement. In view of this, this study attempts to combine HE and bilateral filtering to propose a new method to provide a technical reference for image processing.

3. Image enhancement algorithm method combining HE and bilateral filtering

This part presents a detailed account of the proposed image enhancement algorithm, which combines HE and bilateral filtering. It also outlines the construction and optimization strategy of the algorithm.

In this study, a novel image enhancement algorithm is proposed. This algorithm first utilizes HE technology to enhance image contrast and enhance image details. Next, bilateral filtering technology is used to protect the edge information of the image while effectively suppressing noise. Finally, by introducing an adaptive parameter adjustment strategy, the enhancement effect is automatically adjusted based on the image characteristics to prevent the problem of excessive or insufficient enhancement. This algorithm has shown good performance in improving image visual effects and image quality.

3.1. Image enhancement algorithm integrating HE construction

HE is to obtain a uniform enhanced histogram, and its main idea is to remap the grayscale level by the uniform extension of its probability density function (PDF). Firstly, it constructs a cumulative histogram [16,17], which is normalized to the output image intensity value. Then, it is used as the transfer mapping function. The conversion mainly includes mapping intervals and functions, namely cumulative distribution function (CDF). The PDF and CDF formulas of histograms are used to construct cumulative histograms of images, which are the basis of HE algorithms. By counting the number of pixels at each gray level and normalizing, the cumulative distribution of pixel intensity values can be determined, providing a basis for subsequent gray mapping. The PDF and CDF formulas of the histogram are shown in Eq. (1) [18].

$$\begin{cases} PDF(q) = \dfrac{n_q}{N} \\ CDF(q) = \sum_{i=start}^{q} PDF(i) \end{cases} \tag{1}$$

In Eq. (1), q is the grayscale level of a certain level, n_q is the number of statistical pixels on the n_q grayscale level, N is the total number of pixels in the input image, and N is the minimum grayscale level in a certain interval. The conversion for HE is shown in Eq. (2) [18].

$$F(q) = start + (end - start) \times CDF(q) \tag{2}$$

In Eq. (2), $end - start$ is mapping interval. The HE algorithm and its improved algorithms are divided into histogram segmentation, histogram cropping, reassignment of mapping intervals, and independent equalization. Histogram segmentation is the process of dividing the entire histogram into two or more sub histograms, and then processing each sub histogram separately. Each sub histogram is independently equalized, with the aim of providing a more natural visual effect to the enhanced image. It is used to improve the problem of brightness directional transfer in traditional HE. Histogram clipping is to control the enhancement rate and limit the excessive stretching of certain grayscale levels. As each sub HE is autonomous, it is unable to exceed the specified range, which is the mapping interval. In most cases, for lacking sufficient enhancement regions, the enhancement of narrow sub histograms is not significant. On the other hand, in sub histograms with larger intervals, there may be over enhancement, which can lead to details loss and unnatural output images. The basic steps of HE are shown in Fig. 1 [19].

In Fig. 1, since the histogram distribution range of the image to be enhanced is often small, it is necessary to reassign a new mapping interval to have a reasonable dynamic range. This process is called dynamic range adjustment, and its goal is to improve image brightness contrast appropriately while avoiding details loss caused by excessive enhancement. In the new mapping interval, the brightness values of each pixel will be reassigned, making the brightness distribution more uniform and improving image overall visual effect. In addition, this method also helps to improve image visual balance, so that all parts of the image can be appropriately enhanced, without excessive enhancement in some areas and insufficient enhancement in other areas. There are various quantitative evaluation indicators for image enhancement effects, which can be divided into parameter evaluation indicators and non parameter evaluation indicators. Advanced multi-band excitation (AMBE) is used to measure the performance of different methods in maintaining the original image brightness. It represents the absolute difference between the average brightness of input and output image. AMBE's calculation is shown in Eq. (3) [20].

$$AMBE = |E(I) - E(O)| \tag{3}$$

In Eq. (3), $E(I)$ and $E(O)$ are the average brightness of the input and output images. The quantity of noise introduced by the enhanced image is calculated using Eq. (4), which is applied to an input image $E(O)$ and an output image $O(\mathrm{I},j)$ containing $M \times N$ pixels [20].

$$PSNR = 10\log_{10}\left[\frac{(L-I)^2}{MSE}\right] \tag{4}$$

In Eq. (4), $PSNR$ represents the amount of noise introduced into the enhanced image. The higher the value, the lower the noise introduced in image processing. SSIM is used to measure the images structural similarity, and its calculation is shown in Eq. (5) [21].

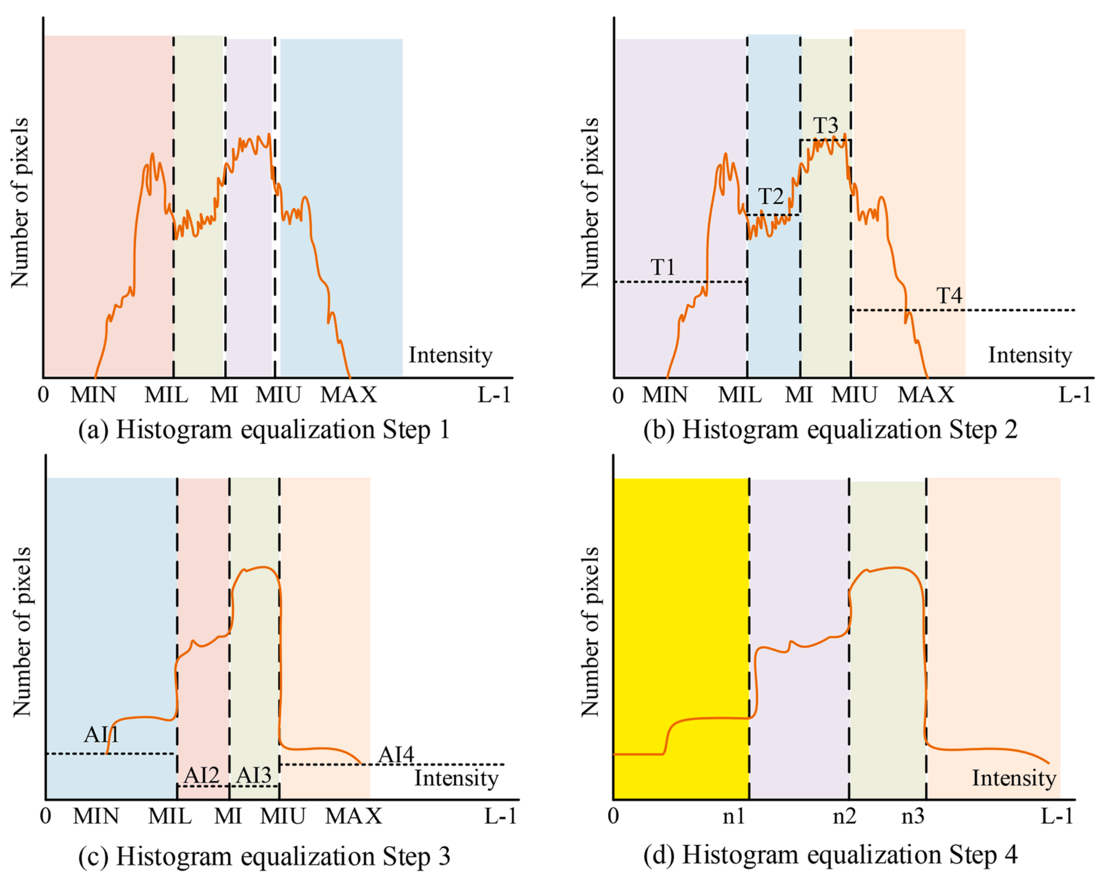

Fig. 1. Basic steps for HE.

$$SSIM(x,y) = \frac{(2u_I u_O + C_1)(2\sigma_{IO} + C_2)}{(u_I^2 + u_O^2 + C_1)(\sigma_I^2 + \sigma_O^2 + C_2)} \quad (5)$$

In Eq. (5), u_I and u_O are the mean values, σ_I and σ_O are the variances, and σ_{IO} is the covariance. C_1 and C_1 are constants. Contrast is used to measure the difference in brightness between the image bright and dark areas. The smaller the difference range, the smaller the contrast, but the greater the contrast, the better. The calculation is shown in Eq. (6) [21].

$$Contrast = \frac{1}{M+N}\sum_{I_{M+N}}(I(x,y) - MB)^2 \quad (6)$$

In Eq. (6), $I(x,y)$ is an image pixel, M and N are the image width and height, and MB is the average brightness of the image I. User Interface Quality Metrics (UIQM) is a non reference evaluation index used to measure the color balance, clarity, and contrast. The calculation is shown in Eq. (7) [22].

$$UIQM = c_1 UICM + c_2 UISM + c_3 UIConM \quad (7)$$

In Eq. (7), $UICM$, $UISM$, and $UIConM$ are the color, clarity, and contrast measurement indexes, c_1, c_2, and c_3 are constants. Combining chromaticity, saturation, and contrast as three components in a linear combination, the measured values are more in line with human visual perception habits. The larger the values, the better the quality of underwater images. The calculation is shown in Eq. (8) [22].

$$UCIQE = e_1\sigma_c + e_2 con_1 + e_3 u_s \quad (8)$$

In Eq. (8), σ_c is the standard deviation of image chromaticity. con_1 is the brightness contrast. u_s is the average value of image saturation. e_1, e_2, and e_3 are the weights of a linear combination. The algorithm process is shown in Fig. 2.

In Fig. 2, when starting the process of image enhancement, the first step is to prepare the image to be enhanced. Next, in order to reduce the complexity of processing, the converted grayscale image's histogram is calculated to reflect image brightness distribution. Whether HE is necessary is determined based on the brightness distribution. If the brightness distribution of the image is already sufficiently uniform, the subsequent HE process can be skipped. If processing is needed, HE is an effective method to enhance image contrast. After processing, the image that has undergone HE is subjected to wavelet decomposition to obtain the low-frequency and high-frequency. Then, further processing and reconstruction are carried out on frequency parts to obtain the enhanced image. Finally, the enhanced image is output, and the entire image enhancement process is completed. Spaceborne LiDAR schematic diagram is shown below [23].

In Fig. 3, the spaceborne LiDAR scanning system is a type of data acquisition system that relies on satellite platforms. It is developed and applied in the late 20th century and continues to develop. This system can actively obtain global surface information and collect large-scale three-dimensional information data.

3.2. Image enhancement optimization by improved bilateral filtering based on HE

When constructing an image enhancement algorithm that incorporates HE and bilateral filtering, the input image is first converted into a grayscale image, and then its histogram is calculated to reflect image brightness distribution. Based on the histogram, determine whether HE is necessary. If necessary, HE is performed to enhance image contrast. Next, the image that has undergone HE is subjected to bilateral filtering, which can remove noise. Finally, the enhanced image is output [24,25]. This algorithm effectively combines the advantages of HE and bilateral filtering, improving image contrast while protecting the edge clarity of the image, thus achieving effective image enhancement. An enhanced version of the Retinex image enhancement algorithm is employed, incorporating an improved bilateral filtering technique. It

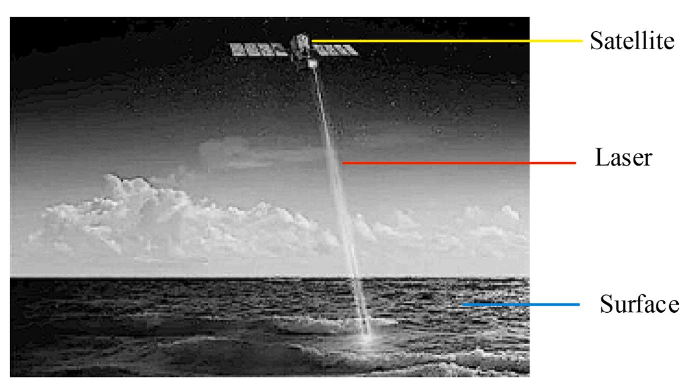

Fig. 3. Schematic diagram of spaceborne LiDAR.

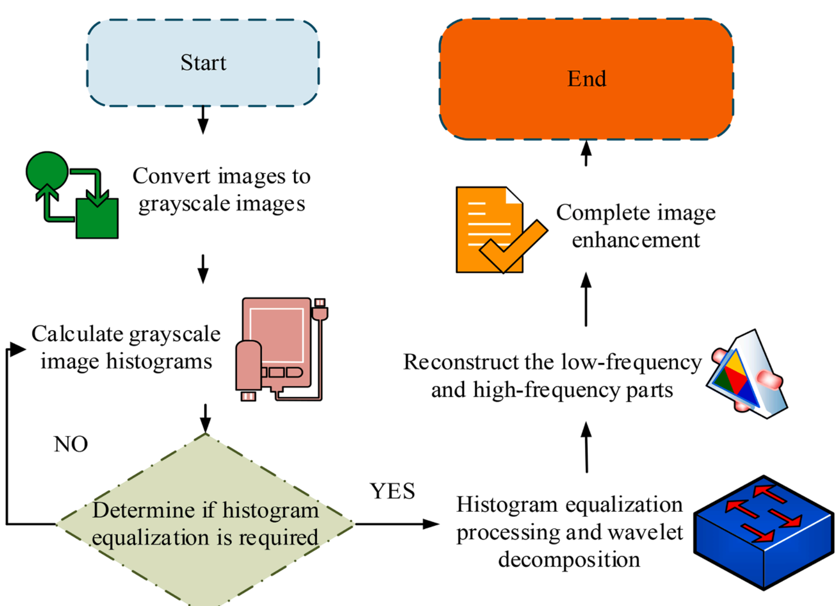

Fig. 2. Basic steps for HE.

specifically focuses on estimating the low-frequency coefficients following wavelet decomposition for illumination images. Through a logarithmic domain conversion, the reflection image is obtained, aiming to eliminate illumination impact on the image. By converting the function to the logarithmic domain, the reflected image is derived after the logarithmic domain conversion process. The logarithmic domain conversion is shown in Eq. (9) [26].

$$R(x,y) = \log_a S(x,y) - k\log_a S(x,y) * F(x,y) \tag{9}$$

In Eq. (9), k is the lighting adjustment parameter and k is the Gaussian kernel function. The bilateral filtering based Retinex (BFR) algorithm is an improvement on the halo phenomenon generated by the single scale Retinex (SSR). It introduces a brightness difference scale parameter in the kernel function. Bilateral filtering smooths the image, as shown in Fig. 4.

In Fig. 4, bilateral filtering, as an efficient and powerful image processing method, not only effectively protects the edge information of the image, but also has a significant effect on eliminating salt and pepper noise. It is also noteworthy that the method is capable of maintaining the clarity of image edges while suppressing noise, which is a crucial aspect in numerous image processing tasks. Especially in applications such as image denoising and enhancement, maintaining the clarity of image edges can often improve the quality of the final result. The expression for bilateral filtering is shown in Eq. (10) [26].

$$I_D(i,j) = \frac{\sum_{k,l} I(k,l)W(i,j,k,l)}{\sum_{k,l} W(i,j,k,l)} \tag{10}$$

In Eq. (10), $I_D(i,j)$ is the output image, $I(k,l)$ is the original image, and $W(i,j,k,l)$ is the kernel function. In flat areas, pixel value change degree is relatively small, and bilateral filtering plays a major role in the spatial domain. In edge area, pixel value change degree is significant. Bilateral filtering plays a role in both the spatial and pixel value domains, effectively avoiding halos caused by significant differences in image brightness at the edges while maintaining image edge information. Improved bilateral filtering is shown in Eq. (11) [27].

$$\begin{cases} W(i,j,k,l) = \left(1 - \sqrt{\dfrac{(i-k)^2 + (j-l)^2}{p}}\right)\exp\left(\dfrac{\| I(i,j) - I(k,l) \|^2}{2\sigma_r^2}\right) \\ \sqrt{(i-k)^2 + (j-l)^2} \le p \end{cases} \tag{11}$$

In Eq. (11), σ_d and σ_r are the distance and brightness difference scale parameters, and p is the filtering window parameter. When the pixels distance between neighborhood and center is below the filtering window parameter, it is determined that pixel position is within the radius range, and its spatial and pixel value domains are effective simultaneously, while maintaining the edge information of the image and eliminating the halo phenomenon. The reflection image obtained after logarithmic domain transformation is beyond 0–230 in pixel value domain, causing phenomena such as graying and whitening of the image, affecting visual effects and subsequent processing. Therefore, it is necessary to stretch the reflected image. Firstly, the maximum and minimum pixel values within the reflected image are calculated. Then, the image is stretched to the range of [0, 230] using the following equation [27].

$$d(x,y) = (R(x,y) - R_{\min}) \times (230 - 0)/(R_{\max} - R_{\min}) \tag{12}$$

In Eq. (12), $d(i,j)$ is the reflected image after stretching, $R(x,y)$ is that before stretching, $R_{\min}$ and $R_{\max}$ are the maximum and minimum grayscale values within $R(x,y)$. The choice of parameters in HE depends mainly on the brightness distribution of the image. By constructing a cumulative histogram and calculating the PDF, the mapping function is determined, and appropriate mapping intervals are selected to achieve global contrast enhancement of the image. The parameter selection in bilateral filtering includes filter window size, illumination adjustment parameters, and brightness difference scale parameters, which are based on the need to effectively suppress noise while preserving image edge information. By analyzing the effect of different mapping intervals on the global contrast of images, the optimal mapping interval is determined to be 0.2. When the window size increases from 3 to 5, the edge preservation ability and noise suppression effect of the image are optimal. In the process of adaptive parameter adjustment, it is essential to conduct a local characteristic analysis of the input image, encompassing aspects such as brightness, contrast, and texture information. The texture complexity of an image is evaluated by calculating its local variance, as shown in Eq. (13).

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N} (I_i - \mu)^2 \tag{13}$$

In Eq. (13), σ^2 represents local variance. N represents the number of pixels in a local area; I_i represents the gray value of the i pixel. μ represents the average gray value of a local area. The initial value of the adaptive parameter should be set, and the window size of the bilateral filter should be adjusted dynamically in accordance with the local characteristics of the image, as illustrated by Eq. (14).

$$p = p_0 + \alpha\cdot\text{texture\_complexity} \tag{14}$$

In Eq. (14), p represents the adjusted window size. p_0 represents the base window size. α represents the adjustment factor. texture_complexity represents the texture complexity index. At the same time, the illumination adjustment parameters are adjusted adaptively, as shown in Eq. (15).

$$k = k_0 \cdot \frac{\text{local\_contrast}}{\text{global\_contrast}} \tag{15}$$

In Eq. (15), k represents the adjusted illumination adjustment parameter. k_0 represents the basic light adjustment parameters. local_contrast represents local contrast. global_contrast represents global contrast. It is possible to achieve a more refined and natural image enhancement by intelligently adjusting the algorithm parameters. The flowchart of the image enhancement algorithm with improved threshold function and bilateral filtering is shown in Fig. 5.

In Fig. 5, first, the image containing noise is processed. Wavelet

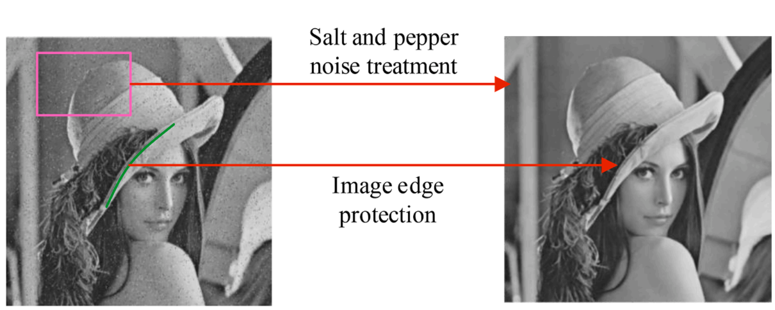

Fig. 4. Comparison before and after bilateral filtering processing.

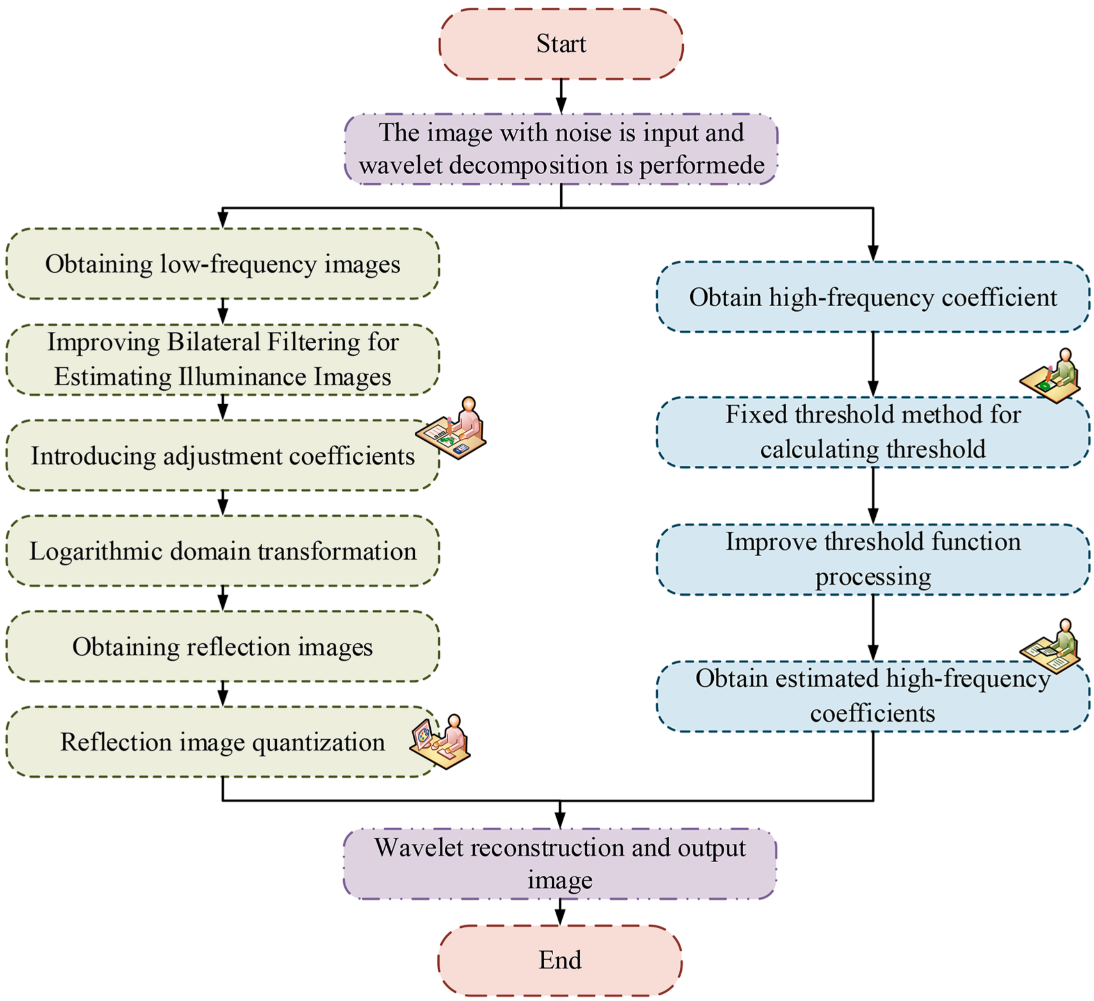

Fig. 5. Schematic diagram of image enhancement algorithm flow for improving threshold function and bilateral filtering.

decomposition is performed using the selected algorithm. Next, the low-frequency image is processed using improved bilateral filtering to estimate the illumination image. Then, lighting adjustment parameters are introduced to obtain a more natural reflection image. Subsequently, the adjusted image is transformed into a logarithmic domain to obtain a reflected image. To further enhance the image contrast and visual effect, HE is conducted on the reflected image. Next, the reflection image that has undergone HE is linearly quantized and stretched to a specified grayscale range to obtain the stretched reflection image. Meanwhile, the high-frequency coefficients obtained from wavelet decomposition are processed using a fixed threshold method to obtain the threshold. Based on this threshold, an improved threshold function re-estimates the wavelet high-frequency coefficients and obtains new high-frequency coefficients. Then, wavelet reconstruction is conducted on the stretched reflection image and the new high-frequency coefficients to obtain the reconstructed image. Finally, the reconstructed image should be output as the final result, thus concluding the entire process. The entire process is clear, and image quality is effectively enhanced through methods HE and bilateral filtering. In order to simplify the implementation process of the research method, it is stipulated that when implemented in a performance limited environment, a basic HE model is constructed by calculating the cumulative histogram of the image and normalizing it. Calculate the cumulative histogram to count the number of pixels at each gray level in the image and calculate their cumulative distribution. The normalization process normalizes the cumulative histogram to the intensity value range of the output image. The mapping function construction uses the CDF as the mapping function to map the gray level of the original image to a new gray level. Simplify the bilateral filtering technique by selecting a fixed-size filtering window during operation to reduce the complexity of parameter selection. Calculate weights based on the spatial distance between pixels and the difference in pixel intensity values. Combining the weights of spatial and intensity domains, filter each pixel to obtain an enhanced image.

4. Image enhancement algorithm optimization analysis combining HE and bilateral filtering

In this part, the proposed image enhancement algorithm was optimized and analyzed. The effectiveness of the algorithm was verified by experiments, and compared with existing methods.

In the optimization analysis of image enhancement algorithms that integrate HE and bilateral filtering, a comparison of images before and after enhancement was first conducted, and an improvement in brightness and contrast was clearly observed. Aiming at the limitations of bilateral filtering, an optimization strategy was proposed, including adjusting filtering parameters and improving weight calculation methods. After implementing the optimization strategy, the effect of bilateral filtering in image enhancement algorithms was more significant, greatly improving the quality of the image.

4.1. Image enhancement algorithm optimization analysis based on HE fusion

In the experiment of optimizing and analyzing image enhancement algorithms for fused HE, the Windows 10 operating system, MATLAB R2020a, and PyCharm IDE were used as experimental platforms. The computer processor was Intel i5–13,400, the graphics processor was AMD 7900XT, the running memory size was 32Gb, the video memory size was 16Gb, and the SSD space size was 500Gb. The parameter settings of the model are shown below.

In Table 1, the Retinex algorithm with improved bilateral filtering was used to estimate low-frequency coefficients in illuminated images. The filtering window size was set to $p = 4$, the illumination image adjustment parameter k was set to 0.6, and the brightness difference scale parameter r was set to 0.002. An improved threshold function processed the high-frequency coefficients, while using coif5 as the wavelet basis function for decomposition and reconstruction. When performing piece-wise linear transformation, the parameters were set to: $a = 4L/8$, $b = 2L/4$, $c = 4L/4$, $d = 4L/4$ ($L = 255$). For the number of decomposition layers, 2 layers were selected. The experimental dataset selected representative CIFAR-10 image dataset and Unsplash dataset for testing and analysis. Experiments were conducted on images with various lighting conditions, including foggy, uneven, and low light. The CIFAR-10 image dataset was used for images with fog and uneven lighting, while the Unsplash dataset was employed for low light and non-real images. The comparative algorithm for the experiment was BFR and the optimization algorithm used. The results of the image enhancement experiment are shown in Fig. 6.

As illustrated in Fig. 6, the application of bilateral filtering resulted in a notable enhancement of contrast in images of varying types. This led to an improved overall recognition of image content, a brighter image, and a more pronounced edge. However, there were still problems such as chromatic aberration, insufficient picture saturation and noise amplification. The overall contrast of the image enhanced by the optimization algorithm was normal, and the picture content was clear. There was basically no obvious image noise, and it had a good visual effect. In order to analyze the enhancement results of each algorithm, the GoPro defugification dataset was selected to test the mean value, variance and information entropy of the algorithm. Moreover, 50 images were selected in the dataset for enhancement. The average results were taken as the analysis results, as shown in Table 2.

In Table 2, the overall image was gray, white, poor contrast and other characteristics. Therefore, it can be observed from the data in the table that the mean value of the original image was high, the variance of the original image was low, and the information entropy was not high. In terms of information entropy, the research method reached 7.70, which was higher than other algorithms. Combined with subjective observation and objective evaluation and analysis, the research method had better enhancement effect.

4.2. Image enhancement algorithm combining HE and optimized bilateral filtering for improving image quality analysis

The bilateral filter used a window size of 4, an illumination image adjustment parameter of 0.6, and a brightness difference scale parameter of 0.002. Wavelet decomposition and reconstruction were carried out at two levels using coif5 as the basis function. The segmented linear transformation parameters were set to $a = 4L/8$, $b = 2L/4$, $c = 4L/4$, $d = 4L/4$ ($L = 255$), and the brightness and contrast of the image were dynamically adjusted. The CIFAR-10 dataset was a well-known benchmark in the fields of machine learning and computer vision, used for testing and analysis. Peak signal-to-noise ratio (PSNR) evaluated image quality and 10 iterations of the algorithm were performed to ensure results robustness and reliability. The reliability of integrating HE and bilateral filtering is shown in Fig. 7.

Table 1
System parameter.

Parameter	Price
Operating system	Windows 10
Software platform	MATLAB R2020a, PyCharm IDE
Filter window size (p)	4
Lighting image adjustment parameters (k)	0.6
Brightness difference scale parameter (r)	0.002
Wavelet basis function	coif5
Piecewise linear transformation parameters (a, b, c, d)	$a = 4L/8$, $b = 2L/4$, $c = 4L/4$, $d = 4L/4$ ($L = 255$)
Decomposition level	2
Data set	CIFAR-10

In Fig. 7, during the initial reliability testing of HE and bilateral filtering, the performance of the filtering strategy using improved HE and the filtering strategy based on bilateral filtering were roughly the same. This was because the dataset used in the two tests was the same, and the actual filtering strategy had little difference in noise processing. However, as the testing progressed, the peak signal-to-noise ratio (PSNR) of the filtering strategy based on improved HE remained slightly higher than that of the filtering strategy based on bilateral filtering. Especially at the end of the testing, the PSNR of the filtering strategy based on bilateral filtering was 0.55, while the PSNR of the filtering strategy based on improved HE was 0.71, with a difference of 7.7 %. The robustness of incorporating HE and bilateral filtering is shown in Fig. 8.

In Fig. 8, the performance of the bilateral filter can be improved through a series of optimizations. The initial bilateral filter had poor performance in handling noise and uncertainty, but by adjusting the window size and image adjustment parameters, its PSNR was improved to 0.75. Further performance enhancement was achieved through wavelet decomposition and reconstruction, with a PSNR of 0.85. When conducting robustness testing on the optimized bilateral filter, the performance of processing complex datasets was significant, with a PSNR of 0.95. Even in complex situations where data noise levels increase or feature correlations change, its performance remained at a high level, with PSNR of 0.85, 0.87, 0.89, and 0.91, all of which were superior to unoptimized bilateral filters. The objective evaluation indicators were compared in complex and changing environments, as shown below.

In Table 3, although the RD algorithm enhanced the results with the highest contrast and UCIQE values, exhibiting high contrast and bright colors, the subjective visual effect was poor, with color distortion caused by excessive enhancement. Its SSIM and PSNR values were low compared to the reference image, indicating low similarity, and overall poor performance. Although the FUSION algorithm exhibited a high degree of similarity with the reference image, it lacked contrast and color information, resulting in a slight decline in performance with respect to these two objective evaluation indicators. The original images were compared in three different scenarios, as shown in Fig. 9.

In Fig. 9, to verify the image enhancement algorithm performance combining HE and bilateral filtering, the method proposed in this chapter was compared with HE and CLAHE image enhancement algorithms. In low illumination conditions, the image quality was not high, the overall appearance was blurry, and the brightness was low. However, when an image enhancement algorithm combining HE and bilateral filtering was applied, the contrast and clarity of the image were significantly improved, although a portion of the image was still dark. In Fig. 9(c), excessive enhancement occurred, resulting in significant color distortion. To further confirm the superiority of the research method, advanced technologies proposed in the past year were selected for comparison. The experiment used an image dataset containing natural landscapes, urban buildings, and low light scenes, and to enhance the validation results, an additional video image dataset was added, totaling 400 images. The selected images were diverse in resolution, lighting conditions and content to ensure the representativeness and universality of the experimental results. The comparison results are shown in Table 4.

As shown in Table 4, the research method achieved a PSNR of 30.5 dB and an SSIM of 0.85 in comparative testing with other advanced methods. Compared to other methods, the research approach also demonstrated better data performance in terms of information entropy

(a) Original drawing (b) After bilateral filtering processing (c) After bilateral filtering processing incorporating histogram equalization

Fig. 6. Comparison of boat and bus image enhancement algorithms.

Table 2
Boat and Bus image enhancement processing results.

Original image and various enhancement algorithms	Mean value	Variance	Information entropy
Original drawing	214.93	23.42	6.40
SSR	157.86	32.77	7.02
BFR	164.30	52.68	7.50
HE	127.53	75.01	5.89
I-R	130.19	46.82	7.49
Algorithm in this article	144.65	60.72	7.70
Original Fig. 2	154.76	55.44	7.13
SSR2	115.70	45.40	7.06
BFR2	206.36	26.36	6.35
HE2	127.67	75.04	5.97
UM2	158.14	56.22	7.29

and UIQM metrics. This indicated that research methods had better performance in improving image details and visual quality. However, the research method's 0.97 in Structural similarity was slightly lower compared to illumination estimation and attention mechanism, indicating that there was still room for improvement in image structure preservation. After statistical analysis, it was found that the research method had an average processing time of 3.21 s per standard 1080p image, which was better than the 4.12 s of HE techniques and the 3.39 s of low light image enhancement. However, the average computational complexity of the research method reached over 40,000 times, which was about 10 % higher than other methods. Compared with other current image enhancement methods, although the proposed algorithm had slightly higher computational complexity, it demonstrated better adaptability in application scenarios that require high real-time performance.

4.3. Discussion

The proposed image enhancement algorithm combined HE and bilateral filtering to improve image quality and reduce image noise. Experimental results showed that the proposed algorithm performed better than existing image enhancement methods on multiple standard datasets. Especially in terms of global contrast enhancement and noise suppression. The optimized HE strategy achieved a PSNR of 0.71, which showed remarkable effect in improving image quality. In addition, the optimal application of the two-sided filtering technique achieved a PSNR of 0.95 on complex data sets, demonstrating its powerful ability in noise suppression. Different image enhancement techniques had different focuses, ranging from HE to illumination estimation and low-light image enhancement based on deep learning. Compared to the

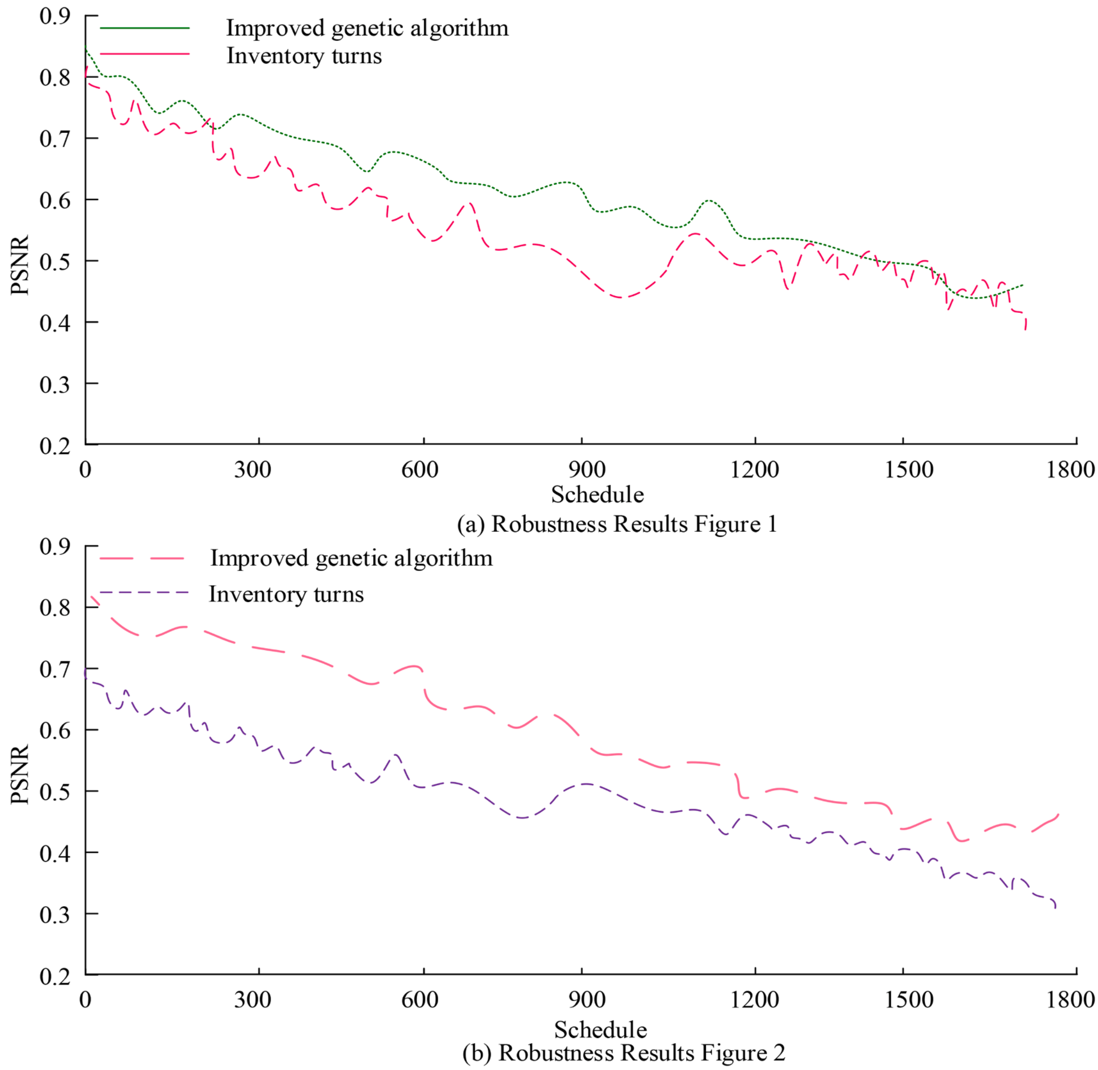

Fig. 7. Reliability of incorporating HE and bilateral filtering.

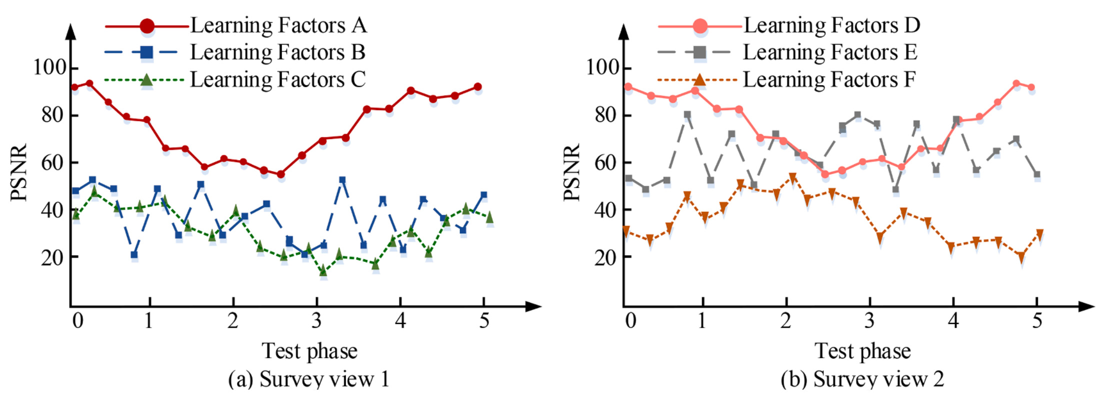

Fig. 8. Peak signal-to-noise ratio of bilateral filtering.

previous research, the research methods used demonstrate superior image detail visibility and more effective noise suppression, while maintaining the natural appearance of the image [33,34]. In particular, when dealing with images under non-uniform illumination and low light conditions, the research method demonstrated its distinctive advantages. The advantage of this research method is that it considers the global and local characteristics of the image in a comprehensive manner, and that it enables the personalization of different types of images through the use of an adaptive parameter adjustment strategy. In addition, the algorithm can still maintain high performance when dealing with complex background and detailed images. Although the research method performs well in many respects, it also have some limitations. For example, when processing images with extreme dynamic range, the algorithm may need to be further adjusted to avoid

Table 3
Objective evaluation indicators comparison of various algorithms in complex and changing environments.

Method	Contrast	Information entropy	PSNR	SSIM	UIQM	UCIQE
Original	60.799	7.498	12.487	0.211	0.389	0.417
Reference	62.844	7.801	Inf	1.000	4.330	0.446
CLAHE	59.470	7.605	14.869	0.304	2.202	0.432
RD	75.117	7.575	15.969	0.433	3.380	0.531
ESIHE	67.673	7.460	16.445	0.469	2.054	0.487
RGHS	66.408	7.147	15.861	0.342	2.347	0.492
FUSION	45.689	7.399	15.350	0.505	2.995	0.417
Our	61.335	7.696	19.207	0.516	3.489	0.463

distortion caused by excessive enhancement. Furthermore, the algorithm exhibits a relatively high computational complexity, which may necessitate the allocation of additional computational resources. This may be a pertinent consideration in the context of real-time image processing applications.

5. Conclusion

One of the core challenges in image processing is to enhance image quality and detail visibility while maintaining the naturalness of the image. A novel image enhancement algorithm that integrates HE and optimized bilateral filtering techniques is designed to address this issue. The research method employs enhanced HE technology to enhance the global contrast of the image. Additionally, it employs bilateral filtering technology to protect the edge information of the image and effectively suppress noise, thereby improving the overall image quality. The introduction of an adaptive parameter adjustment strategy enables the algorithm to dynamically adjust the enhancement effect according to the local characteristics of the image, thus avoiding the problem of excessive enhancement or detail loss that may occur in the traditional method. In the experiment, the research method achieved a PSNR of 0.71 under the improved HE strategy, which showed a significant effect on the improvement of image quality. The PSNR of 0.95 was applied to complex data sets. It proved that the algorithm had remarkable noise suppression ability while protecting image edge information. However, further validation and optimization are needed before this method can be applied in practice. The limitations of the test dataset may prevent the current test results from covering all possible scenarios. Therefore, testing is needed on larger and more complex datasets for robustness and universality verification. Furthermore, the specifications for image quality may differ depending on the specific context or application. It is therefore recommended that future research should make targeted adjustments and optimizations to algorithms, while exploring the applicability of research methods to other special types of images and real-time processing scenarios.

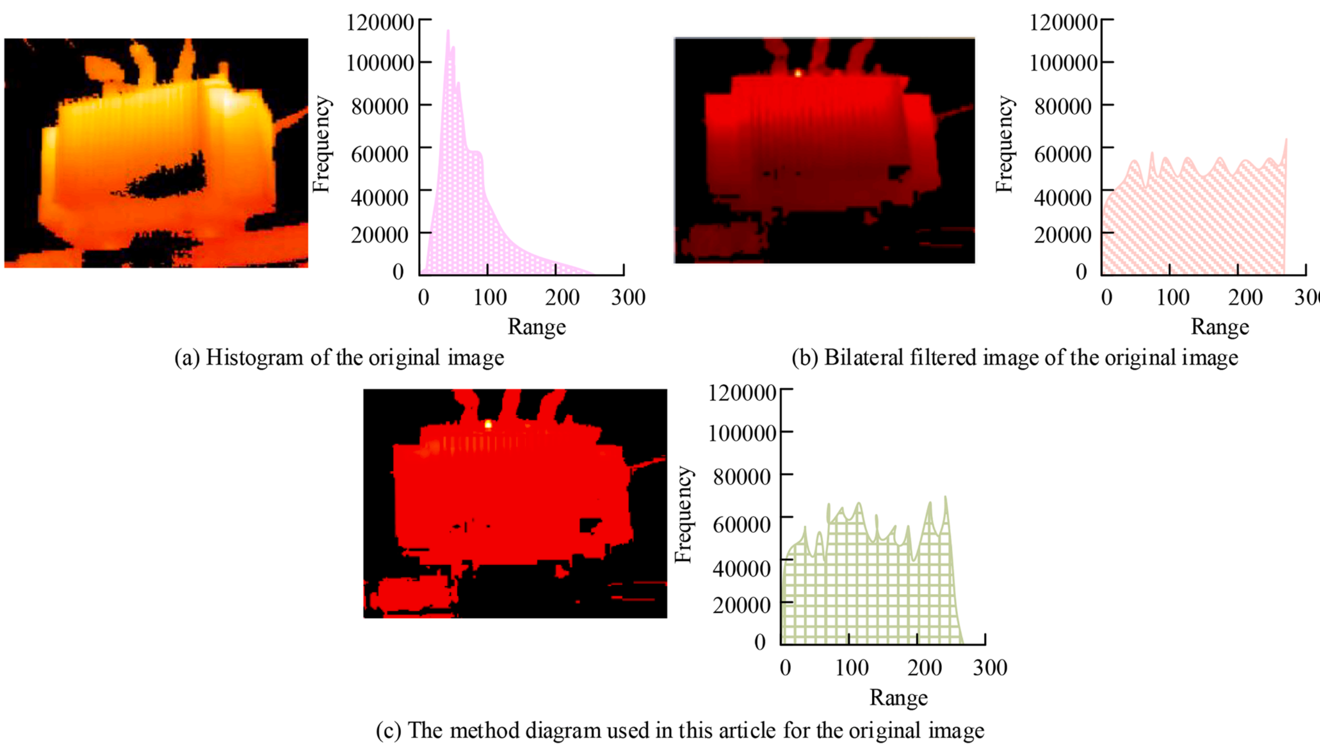

Fig. 9. Comparison of original images in three different scenarios.

Table 4
Comparison of advanced technologies.

Method	Source	PSNR (dB)	SSIM	Information entropy	UIQM	Structural similarity
HE techniques	Roy et al. [28]	29.8	0.79	7.60	0.76	0.93
Illumination estimation	Singh et al. [29]	29.1	0.82	7.20	0.78	0.98
Low-light image enhancement	Wang et al. [30]	28.2	0.83	7.70	0.80	0.94
Gray-scale transformation	Zhou J et al. [31]	30.4	0.81	7.83	0.80	0.94
Attention mechanism	Chen X et al. [32]	30.1	0.82	7.62	0.78	0.98
Research method	\	30.5	0.85	7.95	0.82	0.97

CRediT authorship contribution statement

Mingzhu Wu: Writing – original draft, Methodology, Investigation. **Qiuyan Zhong:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

Both authors declare that there is no conflict of interest in this paper.

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Data availability

No data was used for the research described in the article.

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